Event dashboard > Spec Coding > Frontend > Design System and UI/UX

Design System and UI/UX

A comprehensive design system using AWS Cloudscape ensures consistent, accessible, and delightful user experiences across your chatbot application. This section creates detailed UI/UX specifications leveraging react-design-systems-mcp server guidance that serve as the foundation for all frontend development.

Exercise 1.1: AWS Cloudscape Design System Integration

Create a comprehensive visual design system using AWS Cloudscape with design tokens and style guidelines:

~~~
Create a detailed visual design system for our chatbot application using AWS Cloudscape:

**Use the react-design-systems-mcp server to get AWS Cloudscape best practices and design guidance, then create:**

**AWS Cloudscape Design Tokens Integration**:
```json
{
  "cloudscapeTokens": {
    "colors": {
      "awsui": {
        "colorBackgroundLayoutMain": "#ffffff",
        "colorBackgroundContainerContent": "#ffffff",
        "colorBackgroundContainerHeader": "#fafafa",
        "colorTextAccent": "#0972d3",
        "colorTextBodyDefault": "#16191f",
        "colorTextBodySecondary": "#5f6b7a"
      },
      "semantic": {
        "colorTextStatusSuccess": "#037f0c",
        "colorTextStatusInfo": "#0972d3",
        "colorTextStatusWarning": "#8d6605",
        "colorTextStatusError": "#d91515"
      }
    },
    "secondary": {
      "50": "#f0fdf4",
      "100": "#dcfce7",
      "200": "#bbf7d0",
      "300": "#86efac",
      "400": "#4ade80",
      "500": "#22c55e",
      "600": "#16a34a",
      "700": "#15803d",
      "800": "#166534",
      "900": "#14532d"
    },
    "neutral": {
      "50": "#fafafa",
      "100": "#f5f5f5",
      "200": "#e5e5e5",
      "300": "#d4d4d4",
      "400": "#a3a3a3",
      "500": "#737373",
      "600": "#525252",
      "700": "#404040",
      "800": "#262626",
      "900": "#171717"
    },
    "semantic": {
      "success": "#22c55e",
      "warning": "#f59e0b",
      "error": "#ef4444",
      "info": "#3b82f6"
    }
  }
}
```

**Typography System**:
```json
{
  "typography": {
    "fontFamilies": {
      "primary": "Inter, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif",
      "secondary": "JetBrains Mono, 'Fira Code', Consolas, monospace"
    },
    "fontSizes": {
      "xs": "0.75rem",    // 12px
      "sm": "0.875rem",   // 14px
      "base": "1rem",     // 16px
      "lg": "1.125rem",   // 18px
      "xl": "1.25rem",    // 20px
      "2xl": "1.5rem",    // 24px
      "3xl": "1.875rem",  // 30px
      "4xl": "2.25rem",   // 36px
      "5xl": "3rem"       // 48px
    },
    "fontWeights": {
      "light": 300,
      "normal": 400,
      "medium": 500,
      "semibold": 600,
      "bold": 700
    },
    "lineHeights": {
      "tight": 1.25,
      "normal": 1.5,
      "relaxed": 1.75
    }
  }
}
```

**Spacing and Layout System**:
```json
{
  "spacing": {
    "0": "0",
    "1": "0.25rem",   // 4px
    "2": "0.5rem",    // 8px
    "3": "0.75rem",   // 12px
    "4": "1rem",      // 16px
    "5": "1.25rem",   // 20px
    "6": "1.5rem",    // 24px
    "8": "2rem",      // 32px
    "10": "2.5rem",   // 40px
    "12": "3rem",     // 48px
    "16": "4rem",     // 64px
    "20": "5rem",     // 80px
    "24": "6rem"      // 96px
  },
  "borderRadius": {
    "none": "0",
    "sm": "0.125rem",   // 2px
    "base": "0.25rem",  // 4px
    "md": "0.375rem",   // 6px
    "lg": "0.5rem",     // 8px
    "xl": "0.75rem",    // 12px
    "2xl": "1rem",      // 16px
    "full": "9999px"
  },
  "shadows": {
    "sm": "0 1px 2px 0 rgba(0, 0, 0, 0.05)",
    "base": "0 1px 3px 0 rgba(0, 0, 0, 0.1), 0 1px 2px 0 rgba(0, 0, 0, 0.06)",
    "md": "0 4px 6px -1px rgba(0, 0, 0, 0.1), 0 2px 4px -1px rgba(0, 0, 0, 0.06)",
    "lg": "0 10px 15px -3px rgba(0, 0, 0, 0.1), 0 4px 6px -2px rgba(0, 0, 0, 0.05)",
    "xl": "0 20px 25px -5px rgba(0, 0, 0, 0.1), 0 10px 10px -5px rgba(0, 0, 0, 0.04)"
  }
}
```

Research AWS Cloudscape design system best practices and accessibility color contrast requirements using the react-design-syst
~~~

Expected Outcome

- AWS Cloudscape design token integration with consistent color, typography, and spacing
- Accessible color palette meeting WCAG 2.1 AA contrast requirements following AWS standards
- Scalable typography system using Amazon Ember font with proper hierarchy
- Flexible spacing and layout system consistent with AWS design principles
- Design system documentation aligned with Cloudscape best practices

Exercise 1.2: Component Design Patterns

Create detailed component design patterns and interaction specifications:

~~~
Design comprehensive component patterns for our chatbot interface:

**Chat Interface Components**:

**Message Component Specifications**:
```json
{
  "MessageComponent": {
    "variants": {
      "user": {
        "alignment": "right",
        "backgroundColor": "primary.500",
        "textColor": "white",
        "borderRadius": "lg",
        "maxWidth": "70%",
        "padding": "3"
      },
      "assistant": {
        "alignment": "left",
        "backgroundColor": "neutral.100",
        "textColor": "neutral.900",
        "borderRadius": "lg",
        "maxWidth": "70%",
        "padding": "3"
      },
      "system": {
        "alignment": "center",
        "backgroundColor": "neutral.50",
        "textColor": "neutral.600",
        "borderRadius": "base",
        "maxWidth": "100%",
        "padding": "2"
      }
    },
    "states": {
      "default": "Normal message display",
      "loading": "Typing indicator with animated dots",
      "error": "Error state with retry option",
      "editing": "Inline editing capability"
    },
    "interactions": {
      "hover": "Subtle background color change",
      "select": "Message selection for actions",
      "copy": "Copy message content to clipboard",
      "delete": "Delete message with confirmation"
    }
  }
}
```

**Input Component Specifications**:
```json
{
  "ChatInputComponent": {
    "layout": {
      "container": "Flexible container with send button",
      "textarea": "Auto-expanding textarea with max height",
      "actions": "Send button and optional attachment button",
      "characterCount": "Character counter with limit indication"
    },
    "states": {
      "default": "Ready for input",
      "typing": "User is typing (local state)",
      "sending": "Message being sent (disabled state)",
      "error": "Send failed (retry option)",
      "disabled": "Input disabled (authentication required)"
    },
    "validation": {
      "minLength": 1,
      "maxLength": 4000,
      "emptyMessage": "Cannot send empty messages",
      "tooLongMessage": "Message exceeds character limit"
    },
    "accessibility": {
      "label": "Type your message",
      "placeholder": "Ask me anything...",
      "keyboardShortcuts": "Enter to send, Shift+Enter for new line",
      "screenReader": "Message input field with send button"
    }
  }
}
```

**Navigation and Layout Components**:
```json
{
  "HeaderComponent": {
    "layout": "Fixed header with app branding and user controls",
    "elements": {
      "logo": "App logo with link to home",
      "title": "Chatbot application name",
      "userMenu": "User avatar with dropdown menu",
      "settings": "Settings and preferences access"
    },
    "responsive": {
      "mobile": "Collapsed menu with hamburger icon",
      "tablet": "Condensed layout with essential elements",
      "desktop": "Full layout with all elements visible"
    }
  },
  "SidebarComponent": {
    "purpose": "Conversation history and navigation",
    "elements": {
      "conversationList": "List of recent conversations",
      "newChatButton": "Start new conversation",
      "searchBar": "Search conversation history",
      "settingsLink": "Access to user settings"
    },
    "states": {
      "expanded": "Full sidebar with labels",
      "collapsed": "Icon-only sidebar",
      "hidden": "Completely hidden on mobile"
    }
  }
}
```

**Feedback and Status Components**:
```json
{
  "LoadingComponent": {
    "variants": {
      "spinner": "Circular loading spinner",
      "skeleton": "Skeleton loading for content areas",
      "dots": "Typing indicator with animated dots",
      "progress": "Progress bar for file uploads"
    },
    "accessibility": {
      "ariaLabel": "Loading content",
      "announcements": "Screen reader announcements for state changes"
    }
  },
  "ToastComponent": {
    "types": {
      "success": "Success notifications with checkmark icon",
      "error": "Error notifications with error icon",
      "warning": "Warning notifications with warning icon",
      "info": "Information notifications with info icon"
    },
    "behavior": {
      "duration": "Auto-dismiss after 5 seconds",
      "position": "Top-right corner",
      "stacking": "Multiple toasts stack vertically",
      "actions": "Optional action buttons"
    }
  }
}
```

Research current chat interface design patterns and accessibility best practices using the Web Search MCP server.
~~~

Expected Outcome

- Comprehensive component design patterns for chat interface
- Detailed specifications for message display and input components
- Navigation and layout components with responsive behavior
- Feedback and status components with accessibility considerations
- Interaction patterns and state management for all components

Exercise 1.3: Responsive Design and Layout

Create comprehensive responsive design specifications:

~~~
Design detailed responsive layout specifications for our chatbot application:

**Breakpoint System**:
```json
{
  "breakpoints": {
    "xs": "320px",    // Small mobile devices
    "sm": "640px",    // Large mobile devices
    "md": "768px",    // Tablets
    "lg": "1024px",   // Small desktops
    "xl": "1280px",   // Large desktops
    "2xl": "1536px"   // Extra large screens
  },
  "containerSizes": {
    "xs": "100%",
    "sm": "640px",
    "md": "768px",
    "lg": "1024px",
    "xl": "1280px",
    "2xl": "1536px"
  }
}
```

**Layout Specifications**:

**Mobile Layout (xs, sm)**:
```json
{
  "mobileLayout": {
    "header": {
      "height": "64px",
      "elements": "Logo, hamburger menu, user avatar",
      "behavior": "Fixed position, always visible"
    },
    "sidebar": {
      "default": "Hidden, accessible via hamburger menu",
      "overlay": "Full-screen overlay when opened",
      "animation": "Slide in from left with backdrop"
    },
    "chatArea": {
      "height": "calc(100vh - 64px - 80px)",
      "padding": "16px",
      "scrollBehavior": "Smooth scroll to bottom"
    },
    "inputArea": {
      "height": "80px",
      "position": "Fixed bottom",
      "padding": "16px",
      "background": "White with top border"
    }
  }
}
```

**Tablet Layout (md)**:
```json
{
  "tabletLayout": {
    "header": {
      "height": "72px",
      "elements": "Logo, navigation, user controls",
      "behavior": "Fixed position with shadow"
    },
    "sidebar": {
      "width": "280px",
      "behavior": "Collapsible, persistent when expanded",
      "breakpoint": "Hidden below md, visible above"
    },
    "chatArea": {
      "width": "calc(100% - 280px)",
      "height": "calc(100vh - 72px - 88px)",
      "padding": "24px"
    },
    "inputArea": {
      "height": "88px",
      "padding": "24px",
      "maxWidth": "calc(100% - 280px)"
    }
  }
}
```

**Desktop Layout (lg, xl, 2xl)**:
```json
{
  "desktopLayout": {
    "header": {
      "height": "80px",
      "elements": "Full navigation, search, user controls",
      "behavior": "Fixed with glass morphism effect"
    },
    "sidebar": {
      "width": "320px",
      "behavior": "Always visible, resizable",
      "features": "Conversation search, folders, settings"
    },
    "chatArea": {
      "width": "calc(100% - 320px - 280px)",
      "height": "calc(100vh - 80px - 96px)",
      "maxWidth": "800px",
      "centering": "Centered with max-width constraint"
    },
    "rightPanel": {
      "width": "280px",
      "purpose": "Conversation details, settings",
      "behavior": "Collapsible, context-sensitive"
    },
    "inputArea": {
      "height": "96px",
      "padding": "32px",
      "maxWidth": "800px",
      "centering": "Centered to match chat area"
    }
  }
}
```

**Touch and Interaction Design**:
```json
{
  "touchTargets": {
    "minimumSize": "44px x 44px",
    "recommendedSize": "48px x 48px",
    "spacing": "8px minimum between targets",
    "feedback": "Visual and haptic feedback on touch"
  },
  "gestures": {
    "swipeLeft": "Open sidebar (mobile)",
    "swipeRight": "Close sidebar (mobile)",
    "pullToRefresh": "Refresh conversation (mobile)",
    "longPress": "Message context menu",
    "pinchZoom": "Disabled (prevent accidental zoom)"
  },
  "animations": {
    "duration": "200-300ms for UI transitions",
    "easing": "ease-out for natural feel",
    "reducedMotion": "Respect prefers-reduced-motion setting",
    "performance": "Use transform and opacity for smooth animations"
  }
}
```

**Accessibility Responsive Features**:
```json
{
  "accessibilityFeatures": {
    "focusManagement": {
      "mobile": "Large focus indicators for touch",
      "desktop": "Keyboard navigation with visible focus",
      "skipLinks": "Skip to main content, skip to chat input"
    },
    "textScaling": {
      "support": "Up to 200% text scaling without horizontal scroll",
      "reflow": "Content reflows appropriately at all zoom levels",
      "readability": "Maintains readability at all sizes"
    },
    "colorContrast": {
      "minimum": "4.5:1 for normal text",
      "enhanced": "7:1 for small text",
      "testing": "Automated contrast testing in CI/CD"
    }
  }
}
```

Research current responsive design best practices and mobile-first design principles using the Web Search MCP server.
~~~

Expected Outcome

- Comprehensive responsive design system with mobile-first approach
- Detailed layout specifications for all device categories
- Touch-friendly interaction design with proper target sizes
- Accessibility considerations integrated into responsive design
- Performance-optimized animations and transitions

Exercise 1.4: Accessibility and Inclusive Design

Create comprehensive accessibility specifications following WCAG 2.1 AA guidelines:

~~~
Design detailed accessibility specifications for our chatbot application:

**WCAG 2.1 AA Compliance Framework**:

**Perceivable Requirements**:
```json
{
  "colorAndContrast": {
    "textContrast": {
      "normalText": "4.5:1 minimum ratio",
      "largeText": "3:1 minimum ratio (18pt+ or 14pt+ bold)",
      "nonTextElements": "3:1 for UI components and graphics"
    },
    "colorUsage": {
      "notSoleIndicator": "Color not the only way to convey information",
      "alternatives": "Icons, text labels, patterns for color-blind users",
      "testing": "Simulate color blindness in design process"
    }
  },
  "textAlternatives": {
    "images": "Descriptive alt text for all informative images",
    "decorativeImages": "Empty alt attribute for decorative images",
    "icons": "Accessible names via aria-label or sr-only text",
    "charts": "Text descriptions or data tables for complex graphics"
  },
  "adaptableContent": {
    "semanticMarkup": "Proper HTML5 semantic elements",
    "headingStructure": "Logical heading hierarchy (h1-h6)",
    "landmarks": "ARIA landmarks for page regions",
    "readingOrder": "Logical reading order in DOM"
  }
}
```

**Operable Requirements**:
```json
{
  "keyboardAccessibility": {
    "navigation": {
      "tabOrder": "Logical tab order through all interactive elements",
      "focusIndicators": "Visible focus indicators with 2px minimum outline",
      "skipLinks": "Skip to main content and skip to chat input",
      "shortcuts": "Keyboard shortcuts for common actions"
    },
    "chatInterface": {
      "messageNavigation": "Arrow keys to navigate message history",
      "inputFocus": "Easy return to input field (Escape key)",
      "sendMessage": "Enter to send, Shift+Enter for new line",
      "contextMenus": "Keyboard access to message actions"
    }
  },
  "timingAndMotion": {
    "noTimeouts": "No automatic timeouts for user input",
    "pauseControl": "Ability to pause auto-scrolling animations",
    "reducedMotion": "Respect prefers-reduced-motion setting",
    "seizureProtection": "No flashing content above 3Hz"
  }
}
```

**Understandable Requirements**:
```json
{
  "readableContent": {
    "language": {
      "pageLanguage": "HTML lang attribute set correctly",
      "languageChanges": "Mark language changes with lang attribute",
      "readingLevel": "Clear, simple language appropriate for audience"
    },
    "predictableInterface": {
      "consistentNavigation": "Navigation in same location across pages",
      "consistentIdentification": "Same functionality labeled consistently",
      "contextChanges": "No unexpected context changes on focus/input"
    }
  },
  "inputAssistance": {
    "errorIdentification": "Clear error messages with suggestions",
    "labels": "Descriptive labels for all form inputs",
    "instructions": "Clear instructions for complex interactions",
    "errorPrevention": "Confirmation for destructive actions"
  }
}
```

**Robust Requirements**:
```json
{
  "compatibility": {
    "validMarkup": "Valid HTML5 markup",
    "ariaUsage": "Proper ARIA attributes and roles",
    "assistiveTechnology": "Testing with screen readers",
    "browserSupport": "Works with assistive technology across browsers"
  }
}
```

**Screen Reader Optimization**:
```json
{
  "screenReaderSupport": {
    "chatMessages": {
      "structure": "Each message in list item with proper roles",
      "identification": "Clear identification of sender (user/assistant)",
      "timestamps": "Accessible timestamp information",
      "status": "Live region announcements for new messages"
    },
    "liveRegions": {
      "chatArea": "aria-live='polite' for new messages",
      "status": "aria-live='assertive' for important updates",
      "typing": "Announce when assistant is typing"
    },
    "navigation": {
      "landmarks": "main, navigation, complementary regions",
      "headings": "Proper heading structure for page sections",
      "skipLinks": "Skip to main content and chat input"
    }
  }
}
```

**Inclusive Design Considerations**:
```json
{
  "cognitiveAccessibility": {
    "simplicity": "Clear, simple interface without clutter",
    "consistency": "Consistent patterns and interactions",
    "errorRecovery": "Easy error recovery and undo actions",
    "progress": "Clear progress indicators for multi-step processes"
  },
  "motorAccessibility": {
    "targetSize": "Minimum 44px touch targets",
    "spacing": "Adequate spacing between interactive elements",
    "alternatives": "Alternative input methods (voice, switch)",
    "timeout": "Generous timeouts or no timeouts"
  },
  "visualAccessibility": {
    "zoomSupport": "Support up to 400% zoom without horizontal scroll",
    "highContrast": "Support for high contrast mode",
    "customization": "User preference for font size and colors",
    "darkMode": "Dark mode option for light sensitivity"
  }
}
```

Research current accessibility best practices and assistive technology compatibility using the Web Search MCP server.
~~~

Expected Outcome

- Comprehensive WCAG 2.1 AA compliance framework
- Detailed keyboard navigation and screen reader support
- Inclusive design considerations for diverse user needs
- Accessibility testing procedures and validation methods
- User preference and customization options for accessibility

Exercise 1.5: Design System Documentation and Governance

Create comprehensive design system documentation and governance processes:

~~~
Create detailed design system documentation and governance framework:

**Design System Documentation Structure**:
```json
{
  "documentationSections": {
    "gettingStarted": {
      "overview": "Design system purpose and principles",
      "installation": "How to install and use design tokens",
      "quickStart": "Basic usage examples and patterns"
    },
    "foundations": {
      "colorSystem": "Color palette, usage, and accessibility",
      "typography": "Font system, hierarchy, and usage guidelines",
      "spacing": "Spacing scale and layout principles",
      "iconography": "Icon library and usage guidelines"
    },
    "components": {
      "componentLibrary": "All components with props and examples",
      "usageGuidelines": "When and how to use each component",
      "dosDonts": "Best practices and common mistakes",
      "accessibility": "Accessibility features and requirements"
    },
    "patterns": {
      "layoutPatterns": "Common layout patterns and templates",
      "interactionPatterns": "User interaction flows and behaviors",
      "contentPatterns": "Content structure and writing guidelines"
    }
  }
}
```

**Component Documentation Template**:
```json
{
  "componentDocumentation": {
    "overview": "Component purpose and use cases",
    "anatomy": "Visual breakdown of component parts",
    "props": "All props with types and descriptions",
    "variants": "All visual and behavioral variants",
    "states": "All interactive states (hover, focus, disabled)",
    "examples": "Code examples for common use cases",
    "accessibility": "Accessibility features and requirements",
    "dosDonts": "Usage guidelines and best practices",
    "relatedComponents": "Links to related components"
  }
}
```

**Design Token Management**:
```json
{
  "tokenManagement": {
    "structure": {
      "global": "Base tokens (colors, typography, spacing)",
      "alias": "Semantic tokens (primary, secondary, success)",
      "component": "Component-specific tokens"
    },
    "naming": {
      "convention": "category-property-modifier-state",
      "examples": "color-background-primary-hover",
      "consistency": "Consistent naming across all tokens"
    },
    "distribution": {
      "formats": "CSS custom properties, SCSS variables, JS objects",
      "platforms": "Web, React Native, design tools",
      "automation": "Automated token generation and distribution"
    }
  }
}
```

**Design System Governance**:
```json
{
  "governance": {
    "roles": {
      "maintainers": "Core team responsible for system evolution",
      "contributors": "Designers and developers who propose changes",
      "consumers": "Teams using the design system"
    },
    "processes": {
      "proposalProcess": "RFC process for new components or changes",
      "reviewProcess": "Design and code review requirements",
      "releaseProcess": "Versioning and release management",
      "feedbackProcess": "How to provide feedback and report issues"
    },
    "standards": {
      "designStandards": "Visual design quality standards",
      "codeStandards": "Code quality and testing requirements",
      "documentationStandards": "Documentation completeness requirements",
      "accessibilityStandards": "Accessibility compliance requirements"
    }
  }
}
```

**Quality Assurance and Testing**:
```json
{
  "qualityAssurance": {
    "visualTesting": {
      "tools": "Chromatic for visual regression testing",
      "coverage": "All components and variants tested",
      "automation": "Automated visual testing in CI/CD"
    },
    "accessibilityTesting": {
      "automated": "axe-core integration for automated testing",
      "manual": "Manual testing with screen readers",
      "compliance": "WCAG 2.1 AA compliance validation"
    },
    "performanceTesting": {
      "bundleSize": "Monitor design system bundle size impact",
      "renderPerformance": "Component render performance testing",
      "optimization": "Regular performance optimization reviews"
    }
  }
}
```

**Adoption and Migration Strategy**:
```json
{
  "adoptionStrategy": {
    "training": {
      "workshops": "Design system workshops for teams",
      "documentation": "Comprehensive usage documentation",
      "support": "Office hours and support channels"
    },
    "migration": {
      "assessment": "Audit existing components for migration",
      "planning": "Phased migration approach",
      "tooling": "Automated migration tools where possible",
      "validation": "Post-migration validation and testing"
    },
    "metrics": {
      "adoption": "Track design system component usage",
      "consistency": "Measure design consistency across products",
      "efficiency": "Development time and quality improvements"
    }
  }
}
```

Set up Agent Hooks to automatically validate design system documentation completeness and update component documentation when
~~~

Expected Outcome

- Comprehensive design system documentation with clear usage guidelines
- Component documentation template ensuring consistency across all components
- Design token management system with automated distribution
- Governance processes for design system evolution and quality
- Quality assurance framework with automated testing and validation

Key Deliverables

After completing the Design System and UI/UX exercises, you should have:

1. **Visual Design System**: Comprehensive design tokens, color system, typography, and spacing
2. **Component Design Patterns**: Detailed specifications for all UI components and interactions
3. **Responsive Design System**: Mobile-first responsive layouts for all device categories
4. **Accessibility Framework**: WCAG 2.1 AA compliance with inclusive design considerations
5. **Design System Documentation**: Complete documentation and governance processes

Integration with Advanced Features

Your design system leverages:

- **Web Search MCP Server**: For current design trends and accessibility best practices
- **Agent Steering**: For applying design standards and accessibility requirements
- **Agent Hooks**: For validating design consistency and updating documentation

Next Steps

With your comprehensive design system established, you're ready to translate these specifications into a detailed component architecture. The next section will focus on React component implementation, TypeScript integration, state management patterns, and reusable component libraries that bring your design system to life.

Previous Next

Event ends in 2 days 8 hours 2 minutes.

Ascertain Kiro Workshop

Agentic Coding with Kiro
- Introduction to Kiro
- Setup
- Vibe Coding
- Spec Coding
  - Foundation
  - Infrastructure
  - Frontend
    - Design System and UI/UX
    - Component Architecture
    - Performance Optimization

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